Tokens-per-Parameter Coverage Is Critical for Robust LLM Scaling Law Extrapolation

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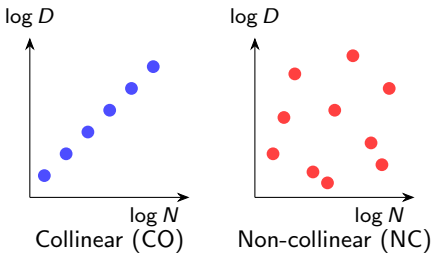






Motivation and collinear designs

- Motivation: LLM runs cost millions; labs fit scaling laws to choose N, D, not trial and error
- Common practice: fixed ratio $k = D/N$, one ray
- On a ray, both terms move together: $L(N, kN) = AN^{-\alpha} + Bk^{-\beta}N^{-\beta} + E$
- Only the gap $\varepsilon = |\alpha - \beta|$ separates A from B



Theory: when the scale pair is identified

- One ray: $\kappa_{A,B} = \Theta(\varepsilon^{-2})$; CI on A is $\Omega(\varepsilon^{-1})\times$ CI on $\psi = A + Bk^{-\alpha}$
- K rays: $\kappa_{A,B} \le \kappa_{\mathrm{target}}$ (leading order) whenever

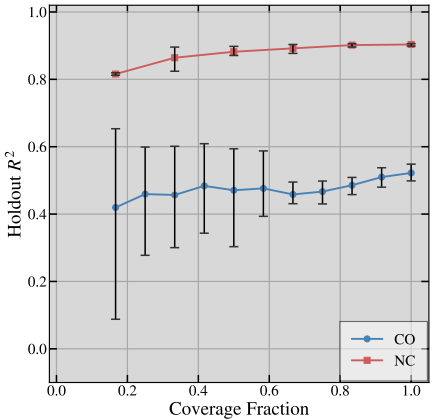
$$\underbrace{\frac{1}{K}\sum_{\ell=1}^{K} k_\ell^{-2\beta_{\mathrm{eff}}} - \left(\frac{1}{K}\sum_{\ell=1}^{K} k_\ell^{-\beta_{\mathrm{eff}}}\right)^2}_{V_K} \;\ge\; \underbrace{\frac{\left(K + \sum_\ell k_\ell^{-2\beta_{\mathrm{eff}}}\right)^2}{K^2\,\kappa_{\mathrm{target}}}}_{\tau_K},$$

- Exact: $\kappa_{A,B} \le \kappa_{\mathrm{target}}$ iff $V_K \ge \tau_K/(1 + 1/\kappa_{\mathrm{target}})^2$
- One ray never passes; two pass once $k_2/k_1 \ge R_{\min}(k_1)$
- Checkable before training

Results: non-collinear designs extrapolate

Comparison (NC vs. CO)	NC wins
Full fan, FP32 (1,460 of 1,500)	97.3%
Full fan, BF16 (296 of 300)	98.7%
Budget-matched, Chinchilla	68.5%
Budget-matched, Droppo-Elibol	57.4%
Budget-matched, Kaplan	80.6%

- 1,900 models, 5M-77M params
- 5 corpora, 4 laws
- Budget-matched: CO subsets below threshold



Holdout R^2 against TPP coverage (Kaplan, RedPajama).

Takeaway and design recipe

- Fit on designs that span TPP ratios
- Recipe: pick k_1, read $R_{\min}(k_1)$, train both ends, split runs evenly
- Check exact $\kappa_{A,B}$ before training
- Code, data, checkpoints: public

  `lab-v2.github.io/tpp-coverage`

